Algebraic Topology

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Contents

1 **Introduction** 1

2 **(Co)homology** 2
- 2.1 Co(chain) complexes . . . 2
- 2.2 Singular (co)homology . . . 4
  - 2.2.1 Basic computations . . . 5
- 2.3 Fundamental properties . . . 7

Lecture 1

1 Introduction

Let X, Y be topological spaces, $f, g : X \to Y$ be continuous functions. Note we will use the terminology 'map := continuous function'. We say that f *is homotopic to* g, written $f \simeq g$, if there exists a map $F : X \times I \to Y$, such that

$$F|_{X\times\{0\}} = f \quad \textit{and} \quad F|_{X\times\{1\}} = g$$

Here $I = [0, 1]$ carries its Euclidean topology. An exercise (sheet 1) is that this defines an equivalence relation on the set of maps $X \to Y$.

Definition 1.1 (homotopy equivalence)

We say that $f : X \to Y$ is a *homotopy equivalence* if there exists a function $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We say that g is a *homotopy inverse* to f.

The same exercise on sheet 1 shows that homotopy equivalence is an equivalence relation on topological spaces.

Example 1.2

1. If $f : X \to Y$ is a homeomorphism, then f is also a homotopy equivalence.
2. $i : \{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivance.
3. The inclusion $i : S^{n-1} \hookrightarrow \mathbb{R}^n \setminus 0$ is a homotopy equivalence.

Definition 1.3

A space X is *contractible* if it is homotopy equivalent to a point.

Idea of Algebraic Topology: Study spaces up to homotopy equivalence. Broadly, we are interested in "connectivity properties" of topological spaces.

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Example 1.4 1. We say a space X is *path-connected* if any two maps $\{*\} \to X$ are homotopic. For example, $\mathbb{R}$ is path connected, $\mathbb{R} \setminus \{0\}$ is not. A corollary of this is the intermediate value theorem.

2. We say a path connected space X is *simply connected* if every map $f : S^1 \to X$ is homotopic to a constant map. Equivalently, every two maps $S^1 \to X$ are homotopic. Or equivalently (again) every continuous map $S^1 \to X$ extends to a continuous map $D^2 \to X$. For example, $\mathbb{R}^2$ is simply connected, by $\mathbb{R}^2 \setminus \{0\}$ is not. If $\gamma : S^1 \to \mathbb{R}^2 \setminus \{0\}$ is continuous, it has a *winding number*, or *degree* $\deg(\gamma) \in \mathbb{Z}$, such that

   - It is invariant under homotopy,
   - If $\gamma_n(t) = \exp(2n\pi it)$, then $\deg(\gamma_n) = n$

   In particular, taking $n = 0$ gives the constant map. Therefore, a constant map has degree 0. A corollary of this is the fundamental theorem of algebra.

3. We say a path connected space X is *k-connected*, if for all $i \leq k$, every map $S^i \to X$ is homotopic to a constant map. In this case, $\mathbb{R}^n$ is $(n-1)$-connected, but $\mathbb{R}^n \setminus 0$ is not. More precisely, maps from $S^{n-1} \to \mathbb{R}^n \setminus 0$ have a degree $\deg(f) \in Z$, such that degree is a homotopy invariant, and the degree of the inclusion map is 1, and the degree of the constant map is 0. A corollary of this is the Brouwer fixed point theorem.

Lecture 2

2 (Co)homology

2.1 Co(chain) complexes

We will define invariants of topological spaces in two steps:

(a) We associate to X a (co)chain complex,

(b) We take the (co)homology of the complex

The topology enters in step (a), whereas (b) is just homological algebra.

Definition 2.1 (chain complex)

A *chain complex*@ (C_*, ∂) consists of a sequence $\{C_i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial_n : C_n \to C_{n-1}$ for all n, satisfying $\partial_n \circ \partial_{n+1} = 0$. We also write this conditions as $\partial^2 = 0$.

Definition 2.2 (homology group)

The *i-th homology group* of a chain complex (C_*, ∂) is

$$\mathrm{H}_i(C_*, \partial) = \frac{\ker(\partial_i : C_i \to C_{i-1})}{\mathrm{im}(\partial_{i+1} : C_{i+1} \to C_i)}$$

We write

$$\mathrm{H}_*(C_*, \partial) = \bigoplus_{i\in\mathbb{Z}} \mathrm{H}_i(C_*, \partial)$$

Definition 2.3 (cochain complex)

A *cochain complex* (C^*, ∂) consists of a sequence $\{C^i\}_{i\in\mathbb{Z}}$ of abelian groups, along with *boundary homomorphisms* $\partial^n : C^n \to C^{n+1}$ for all n, satisfying $\partial^{n+1} \circ \partial^n = 0$. We also write this conditions as $\partial^2 = 0$.

Definition 2.4 (cohomology group)

The *i-th cohomology group* of a cochain complex (C^*, ∂) is

$$\mathrm{H}^i(C^*, \partial) = \frac{\ker(\partial^i : C^i \to C^{i+1})}{\mathrm{im}(\partial^{i-1} : C^{i-1} \to C^i)}$$

We write

$$\mathrm{H}^*(C^*, \partial) = \bigoplus_{i \in \mathbb{Z}} \mathrm{H}_i(C^*, \partial)$$

Notation 2.5. Elements of $\ker(\partial)$ are called *cycles* (in a chain complex), *cocycles* (in a cochain complex). Elements of $\mathrm{im}(\partial)$ are called *boundaries* (and *coboundary* resp.).

Elements of H_* are called *homology classes*, and elements of H^* are called *cohomology classes*.

We will call ∂ the *differential*.

Definition 2.6 (chain map)

Given chain complex (C_*, ∂), (D_*, ∂), a *chain map* $f : C_* \to D_*$ comprises group homomorphism $f_i : C_i \to D_i$, such that

$$\begin{array}{ccc} C_i & \xrightarrow{f_i} & D_i \\ \downarrow{\scriptstyle \partial} & & \downarrow{\scriptstyle \partial} \\ C_{i-1} & \xrightarrow[f_{i-1}]{} & D_{i-1} \end{array}$$

commutes.

Lemma 2.7. A chain map $f : C_* \to D_*$ induces homomorphisms $f_* : \mathrm{H}_i(C_*) \to \mathrm{H}_i(D_*)$.

Proof. Let $a \in \mathrm{H}_i(C_*)$, and we can choose a cycle $\alpha \in C_i(C_*)$, with $\partial\alpha = 0$. In this case,

$$\partial f_i(\alpha) = f_{i-1}(\partial\alpha) = f_{i-1}(0) = 0$$

So $f_i(\alpha) \in D_i$ is a cycle. We set $f_*(a) = [f_i(\alpha)] \in \mathrm{H}_i(D_*)$. Next, we need to show that this is independent of choices. Suppose $a = [\alpha] = [\alpha']$. Then $\alpha - \alpha' \in \mathrm{im}(\partial)$. Therefore, we can write $\alpha - \alpha' = \partial\tau$. In this case,

$$f_i(\alpha) - f_i(\alpha') = f_i(\partial\tau) = \partial f_{i+1}(\tau) \in \mathrm{im}(\partial)$$

So we conclude $[f_i(\alpha)] = [f_i(\alpha')] \in \mathrm{H}_i(D_*)$. It is clear that f_* defines a homomorphism. □

Correspondingly, we can define *cochain maps* $C^* \to D^*$, that is, $f^i : C^i \to D^i$ such that

$$\begin{array}{ccc} C^{i+1} & \xrightarrow{f^{i+1}} & D^{i+1} \\ \uparrow{\scriptstyle \partial} & & \uparrow{\scriptstyle \partial} \\ C^i & \xrightarrow[f^i]{} & D^i \end{array}$$

commutes.

Exercise: This construction is functorial. That is, $(\mathrm{id}_{C_*})_* = \mathrm{id}_{\mathrm{H}_*}$, and if we have $f : C_* \to D_*$, $g : D_* \to E_*$, then $(g \circ f)_* = g_* \circ f_*$.

Goal: To associate to a topological space X (co)chain complex $C_*(X)$ and $C^*(X)$. The definition works for any X, but the theories are better behaved for "nicer" X.

2.2 Singular (co)homology

We will develop singular (co)homology.

Definition 2.8 (standard simplex, face)

The *standard simplex* is

$$\Delta^n = \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid t_i \geq 0, \sum_i t_i = 1 \right\}$$

The i-th face Δ^n is

$$\Delta^n_i = \Delta^n \cap \{t_i = 0\}$$

and we have a canonical homeomorphism

$$\begin{aligned} \delta_i : \Delta^{n-1} &\to \Delta^{n-1}_i \\ (t_0, \dots, t_{n-1}) &\mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}) \end{aligned}$$

Definition 2.9 (singular simplex, singular chain complex)

If X is a topological space, a *singular n-simplex* in X is a continuous map $\sigma : \Delta^n \to X$.

The *singular chain complex* is given by the the free abelian groups

$$C_i(X) = \bigoplus_{\sigma \text{ singular } n\text{-simplex}} \mathbb{Z}\sigma$$

We define the boundary map $C_i(X) \to C_{i-1}(X)$ via

$$\partial\sigma = \sum_{j=0}^{i} (-1)^j \sigma|_{\Delta^i_j}$$

where

$$\sigma|_{\Delta^i_j} := \sigma \circ \delta_j : \Delta^{n-1} \to X$$

Note if $\{v_i\}_{i=0}^n$ are $n+1$ ordered points in $\mathbb{R}^{n+1}$, and if $\{v_i - v_0\}_{i=1}^n$ are linearly independent, then the convex hull of $v_0, \dots, v_n$ is an n-simplex, which we will call

$$[v_0, \dots, v_n]$$

which is given by the map

$$\begin{aligned} \Delta^n &\to \mathbb{R}^{n+1} \\ t &\mapsto \sum_i t_i v_i \end{aligned}$$

We orient the edges of the standard simplex (and thus any simplex $[v_0, \dots, v_n]$) by saying that $v_i < v_j$ if $i < j$.

More concretely, for $n = 2$, we have the oriented edges $v_0 \to v_1, v_0 \to v_2, v_1 \to v_2$.

Lemma 2.10. $\partial^2 = 0$.

Proof. If σ is an n-simplex, defined on $[v_0, \dots, v_n]$, then

$$\partial\sigma = \sum_j (-1)^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, v_n]}$$

and so

$$\partial(\partial\sigma) = \sum_{i<j} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \widehat{v_i}, \dots, \widehat{v_j}, \dots, v_n]} + \sum_{i>j} (-1)^{i+1} - 1^j \sigma|_{[v_0, \dots, \widehat{v_j}, \dots, \widehat{v_i}, \dots, v_n]} = 0$$

since the signs cancel. □

Definition 2.11

The *singular homology groups of* X, $\mathrm{H}_i(X)$, are the homology groups of the singular chain complex.

Remark 2.12. Since all we used is the topology on X, $\mathrm{H}_i(X)$ is clearly a topological (read homeomorphism) invariant of X.

Moreover, if $f : X \to Y$ is continuous, $\sigma : \Delta^n \to X$ is an n-simplex on X, then $f \circ \sigma$ is an n-simplex on Y. The fact that

$$(f \circ \sigma) \circ \delta_j = f \circ (\sigma \circ \delta_j)$$

shows that f induces a chain map.

Lecture 3

Definition 2.13 (singular cochain complex)

The *singular cochain complex* of X has

$$C^i(X) = \mathrm{Hom}_{\mathbb{Z}}(C_i(X), \mathbb{Z})$$

and the boundary operator

$$\partial^* \psi(\sigma) = \psi(\partial \sigma)$$

for $\psi \in C^i(X)$, $\sigma \in C_{i+1}(X)$. That is, it is adjoint to $\partial : C_i(X) \to C_{i-1}(X)$.

Since $\partial^2 = 0, (\partial^*)^2 = 0$. The *singular cohomology* is

$$\mathrm{H}^*(X) = \bigoplus_{i \geq 0} \mathrm{H}^i(C^*, \partial^*)$$

If $f : X \to Y$ continuous, then f induces a chain map $f_* : C_*(X) \to C_*(Y)$ via $f_*(\sigma) = f \circ \sigma$, satisfying $\partial f_* = f_* \partial$. Similarly, $f : X \to Y$ defines a *pullback map* $f^* : C^*(Y) \to C^*(X)$, given by

$$f^*(\psi)(\tau) = \psi(f_*(\tau))$$

Again, we have that

$$\partial^* f^*(\psi)(\tau) = f^* \psi(\partial \tau) = \psi(f_* \partial \tau) = \psi(\partial f_* \tau) = (\partial^* \psi)(f_* \tau) = f^*(\partial^* \psi)(\tau)$$

So f^* is a cochain map, i.e. $\partial^* f^* = f^* \partial^*$. Therefore, we have an induced homomorphism on cohomology,

$$f^* : \mathrm{H}^*(Y) \to \mathrm{H}^*(X)$$

Warning: by definition,

$$C^i(X) = \mathrm{Hom}_{\mathbb{Z}}(C_i(X), \mathbb{Z})$$

but it is not true (in general) that

$$\mathrm{H}^i(X) = \mathrm{Hom}_{\mathbb{Z}}(\mathrm{H}_i(X), \mathbb{Z})$$

2.2.1 Basic computations

Lemma 2.14 (homology of a point)**.** Let $X = \{\bullet\}$. Then

$$\mathrm{H}_n(X) = \begin{cases} \mathbb{Z} & n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof. By definition, $C_i(X)$ is the free abelian groups on continuous map $\Delta^i \to \{\bullet\}$. But there is only one such map, namely the constant map (for $i \geq 0$). Hence the chain complex looks like

$$\cdots \longrightarrow C_2(X) \longrightarrow C_1(X) \longrightarrow C_0(X) \longrightarrow 0$$

where each $C_i(X)$ is $\mathbb{Z}$. Computing the boundary maps,

$$\partial(\sigma_n) = \begin{cases} \sigma_{n-1} & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

Hence the chain complex is

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z} \xrightarrow{0} 0$$

We can then compute the homology groups explicity. □

Lemma 2.15. If we write

$$X = \bigsqcup_{\alpha \in A} X_\alpha$$

as a disjoint union of path components, then

$$\mathrm{H}_i(X) = \bigoplus_{\alpha \in A} \mathrm{H}_i(X_\alpha)$$

Moreover, if X is path connected (and nonempty), then $\mathrm{H}_0(X) = \mathbb{Z}$.

Proof. Since Δ^i is path connected, the image of a continuous map $\sigma : \Delta^i \to X$ must lie within some X_α. Moreover, so does the image of all of the boundary faces $\sigma \circ \delta_j$. So in fact

$$(C_i(X), \partial) = \left(\bigoplus_{\alpha \in A} C_i(X_\alpha), \partial \right)$$

Note that a general element of $C_i(X)$ is a finite linear combination of i-simplicies in X. So its image can only meet finitely many of the X_α.

Now suppose X is path connected. Define a function

$$\begin{aligned} \varepsilon : C_0(X) &\to \mathbb{Z} \\ \sum n_i \sigma_i &\mapsto \sum n_i \end{aligned}$$

Since X is nonempty, ε is surjective. On the other hand, if $\tau \in C_1(X)$, then $\varepsilon(\tau(1) - \tau(0)) = 0$. So by linearity,

$$\mathrm{im}(\partial : C_1(X) \to C_0(X)) \subseteq \ker(\varepsilon)$$

Recall

$$\mathrm{H}_0(X) = \frac{C_0(X)}{\mathrm{im}(\partial : C_1(X) \to C_0(X))}$$

and so ε descends to a map $\mathrm{H}_0(X) \to \mathbb{Z}$. Now we will use the fact that X is path connected. Suppose

$$\sum n_i \sigma_i \in \ker(\varepsilon)$$

and fix a base point $p \in X$. Since the σ_i are 0-simplicies, they correspond to points in X. Moreover, since X is path connected, we can choose paths

$$\tau_i : \Delta^1 \to X$$

such that $\tau_i(1) = \sigma_i$ and $\tau_i(0) = p$. In this case,

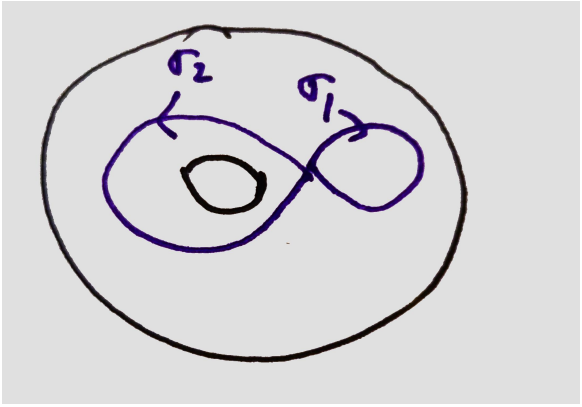
$$\partial \left(\sum n_i \tau_i \right) = \sum n_i \partial \tau_i = \sum n_i \sigma_i - (\sum n_i) p$$

But we assumed $\sum n_i = 0$, and so $\ker(\varepsilon) \subseteq \mathrm{im}(\partial : C_1(X) \to C_0(X))$. This then gives the required result, as the induced map

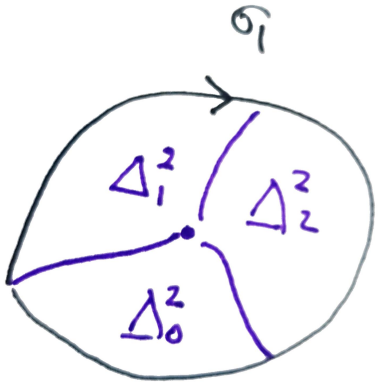
$$\mathrm{H}_0(X) \to \mathbb{Z}$$

is then an isomorphism. □

Informal conjecture: We can't compute anything else.
Therefore, we need to develop some more structure about (co)homology.
What are we doing? Consider the case of X being an annulus in $\mathbb{R}^2$. We have 1-simplices



(σ_1 null-homotopic, σ_2 winds around the origin)
In this case, $\partial\sigma_1 = \partial\sigma_2 = 0$, and so they define homology classes.



Moreover, $[\partial\sigma_1] = 0$, since we have triangles τ_0, τ_1, τ_2 such that

$$\partial(\tau_0 + \tau_1 + \tau_2) = \sigma_1$$

From this intuition, $[\sigma_2] \neq 0$. But to show even this, we need some structure theorems.

Lecture 4

2.3 Fundamental properties

(Co)homology is useful by virtue of various structural properties.

Theorem 2.16 (homotopy invariance)**.** If $f, g : X \to Y$ are homotopic, then the induced maps f_*, g_* on homology, and the induced maps f^*, g^* on cohomology agree.

Corollary 2.17. If $X \simeq Y$, then $H_*(X) \simeq Y$, and if f is a homotopy equivalence, then f_* induces the isomorphism. For cohomology we have the similar statement, but with f^*.

Proof of the corollary. $X \simeq Y$ via f is saying that there exists $g : Y \to X$, such that $g \circ f \simeq \mathrm{id}_X$, and $f \circ g \simeq \mathrm{id}_Y$. That is,

$$g_* \circ f_* = \mathrm{id} \quad \text{and} \quad f_* \circ g_* = \mathrm{id}$$

□

We think of this as saying that (co)homology is "insensitive to inessential deformations".

Example 2.18

Recall $\{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivalence, and so

$$\mathrm{H}_*(\mathbb{R}^n) = \begin{cases} \mathbb{Z} & * = 0 \\ 0 & \text{otherwise} \end{cases}$$

The other key structural property relies on some homological algebra.

Definition 2.19 (exact sequence)

An *exact sequence* is a chain complex with $H_*(C_*, \partial) = 0$. That is,

$$\mathrm{im}(\partial : C_{n+1} \to C_n) = \ker(\partial : C_n \to C_{n-1})$$

Similarly, a cochain complex is *exact* if $H^*(C^*, \partial) = 0$.
If A, B, C are abelian groups, and we have

$$A \xrightarrow{\alpha} B \xrightarrow{\beta} C$$

Then this sequence is *exact at* B if $\ker(\beta) = \mathrm{im}(\alpha)$.

Definition 2.20 (short exact sequence)

A *short exact sequence* is an exact sequence of the form

$$0 \longrightarrow A \xrightarrow{\alpha} B \xrightarrow{\beta} C \longrightarrow 0$$

which is exact at all places. That is, α is injective, β is surjective, and $\mathrm{im}(\alpha) = \ker(\beta)$.

Example 2.21 (i) If

$$0 \longrightarrow A \xrightarrow{\alpha} B \longrightarrow 0$$

is exact, then α is an isomorphism.

(ii) If

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\alpha} G \xrightarrow{\beta} \mathbb{Z}/n \longrightarrow 0$$

is exact, then we could have $G = \mathbb{Z} \oplus \mathbb{Z}/n$, or we could have $G = \mathbb{Z}$, with $\alpha(x) = nx$.

Theorem 2.22 (Mayer-Vietoris). If $X = A \cup B$, with A, B open subsets of X. Then there exists *Mayer-Vietoris boundary homomorphism*

$$\partial_{MV} : \mathrm{H}_{i+1}(X) \to \mathrm{H}_i(A \cap B)$$

for all i, so that the sequence

$$\cdots \longrightarrow \mathrm{H}_{i+1}(X) \xrightarrow{\partial_{MV}} \mathrm{H}_i(A \cap B) \xrightarrow{i_{A*} \oplus i_{B*}} \mathrm{H}_i(A) \oplus \mathrm{H}_i(B) \xrightarrow{j_{A*} - j_{B*}} \mathrm{H}_i(X) \qquad \cdots$$

is exact, where

$$\begin{array}{ccc} A \cap B & \xrightarrow{i_A} & A \\ {\scriptstyle i_B}\downarrow & & \downarrow{\scriptstyle j_A} \\ B & \xrightarrow[j_B]{} & X \end{array}$$

Similarly, we have a homomorphism

$$\partial^*_{MV} : H^i(A \cap B) \to H^{i+1}(X)$$

which makes the sequence

$$\cdots \longrightarrow \mathrm{H}^{i-1}(A \cap B) \xrightarrow{\partial^*_{MV}} \mathrm{H}^i(X) \xrightarrow{j_A^* \oplus -j_B^*} \mathrm{H}^i(A) \oplus \mathrm{H}^i(B) \xrightarrow{i_A^* + i_B^*} \mathrm{H}^i(A \cap B) \xrightarrow{\partial^*_{MV}} \mathrm{H}^{i-1}(X) \longrightarrow \cdots$$

exact.

Remark 2.23. (i) The Mayer-Vietoris maps are not induced by maps of spaces, but they are constructed algebraically. Suppose $\sigma \in C_{i+1}(X)$ is a cycle, and suppose we can write $\sigma = \sigma_A + \sigma_B$, where $\sigma_A \in C_{i+1}(A)$ and $\sigma_B \in C_{i+1}(B)$ are chains, and in general, not cycles. Since $\partial\sigma = 0, \partial\sigma_A + \partial\sigma_B = 0$. We define

$$\partial_{MV}(\sigma) = [\partial\sigma_A] \in \mathrm{H}_i(A \cap B)$$

Since $\partial\sigma_A \in C_i(A)$, we must have that $\partial\sigma_A = -\partial\sigma_B \in C_i(A \cap B)$. As $\partial^2 = 0$, $\partial\sigma_A$ is closed and so it represents a class in homology.

This is (correct) intuition, but not a proof.

(ii) The Mayer-Vietoris sequence is natural for maps of pairs. That is, if $X = A \cup B$, $Y = C \cup D$, and $f : X \to Y$ has $f(A) \subseteq C$, $f(B) \subseteq D$, then we have a map of long exact sequences

$$\begin{array}{ccccccccccc} \cdots \longrightarrow & \mathrm{H}_{i+1}(X) & \longrightarrow & \mathrm{H}_i(A \cap B) & \longrightarrow & \mathrm{H}_i(A) \oplus \mathrm{H}_i(B) & \longrightarrow & \mathrm{H}_i(X) & \longrightarrow & \mathrm{H}_{i-1}(A \cap B) & \longrightarrow \cdots \\ & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_* \oplus f_*} & & \downarrow{\scriptstyle f_*} & & \downarrow{\scriptstyle f_*} & \\ \cdots \longrightarrow & \mathrm{H}_{i+1}(Y) & \longrightarrow & \mathrm{H}_i(C \cap D) & \longrightarrow & \mathrm{H}_i(C) \oplus \mathrm{H}_i(D) & \longrightarrow & \mathrm{H}_i(Y) & \longrightarrow & \mathrm{H}_{i-1}(C \cap D) & \longrightarrow \cdots \end{array}$$

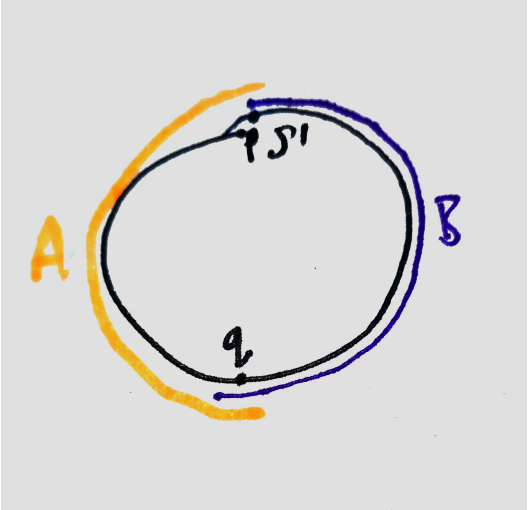
where all the squares commute.

We will prove these later when we use them.

Example 2.24

$$\mathrm{H}_*(S^1) = \begin{cases} \mathbb{Z} & * = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof. Here, $S^1 = A \cup B$, where $A, B \simeq \{*\}$ and $A \cap B \simeq \{p, q\}$.



Using this, we know $H_*(A), H_*(B), H_*(A \cap B)$. For $i \geq 2$, we have

$$\cdots \longrightarrow \underbrace{H_i(A) \oplus H_i(B)}_{=0} \longrightarrow H_i(X) \longrightarrow \underbrace{H_{i-1}(A \cap B)}_{=0} \longrightarrow \cdots$$

and so $H_i(X) = 0$ for all $i \geq 2$. For lowe degrees, we have

$$\begin{array}{ccccccc} \underbrace{H_1(A \cap B)}_{=0} & \longrightarrow & \underbrace{H_1(A) \oplus H_1(B)}_{=0} & \longrightarrow & H_1(X) & & \\ & & & \overset{\partial_{MV}}{\searrow} & & & \\ \underbrace{H_0(A \cap B)}_{\langle p \rangle \oplus \langle q \rangle} & \xrightarrow{\alpha} & \underbrace{H_0(A) \oplus H_0(B)}_{=\mathbb{Z} \oplus \mathbb{Z}} & \longrightarrow & \underbrace{H_0(X)}_{=\mathbb{Z}} & \longrightarrow & 0 \end{array}$$

where $\alpha(n, m) = (n + m, n + m)$. Moreover, we use that S^1 is path connected, and H_0 is generated by any point. Therefore, $H_1(X)$ sits in an exact sequence

$$0 \longrightarrow H_1(X) \xrightarrow{\partial_{MV}} \mathbb{Z}^2 \xrightarrow{\alpha} \mathbb{Z}^2 \xrightarrow{\beta} \mathbb{Z} \longrightarrow 0$$

and so,

$$H_1(X) \simeq \ker(\alpha) = \langle (1, -1) \rangle = \mathbb{Z}(p - q)$$

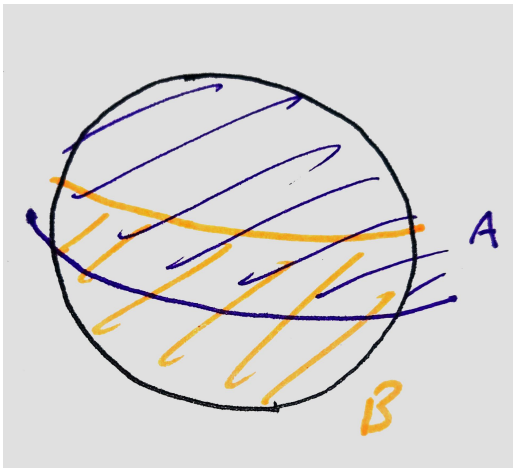
□

Example 2.25

$$H^*(S^n) = H_*(S^n) = \begin{cases} \mathbb{Z} & * = 0, n \\ 0 & \text{otherwise} \end{cases}$$

Lecture 5

Proof. We will compute the cohomology. Define $A, B \subseteq S^n$ by



Then A, B are contractible, and $A \cap B$ is homotopic to S^{n-1}. We work inductively in n, using the Mayer-Vietoris sequence. The sequence gives

$$\begin{array}{ccccc} \cdots & \longrightarrow & \mathrm{H}^{i-1}(S^n) & & \\ & \overset{\partial^*_{MV}}{\swarrow} & & & \\ \mathrm{H}^i(S^n) & \longrightarrow & \mathrm{H}^i(*) \oplus \mathrm{H}^i(*) & \longrightarrow & \mathrm{H}^{i-1}(S^{n-1}) \\ & \overset{\partial^*_{MV}}{\swarrow} & & & \\ \mathrm{H}^{i+1}(S^n) & \longrightarrow & \mathrm{H}^{i+1}(*) \oplus \mathrm{H}^i(*) & \longrightarrow & \cdots \end{array}$$

If $i > 0$, we have

$$0 \longrightarrow \mathrm{H}^i(S^{n-1}) \xrightarrow{\partial^*_{MV}} \mathrm{H}^{i+1}(S^n) \longrightarrow 0$$

and so

$$\mathrm{H}^i(S^{n-1}) = \mathrm{H}^{i+1}(S^n)$$

This gives us almost everything inductively. We can assume $n \geq 2$. At the bottom of the sequence, we have

$$\underbrace{\mathrm{H}^0(S^n)}_{=\mathbb{Z}} \longrightarrow \underbrace{\mathrm{H}^0(*) \oplus \mathrm{H}^0(*)}_{=\mathbb{Z}\oplus\mathbb{Z}} \xrightarrow{\alpha} \underbrace{\mathrm{H}^0(S^{n-1})}_{=\mathbb{Z}} \longrightarrow \mathrm{H}^1(S^n) \longrightarrow 0$$

We have shown (on examples sheet 1) that $\mathrm{H}^0(X) = 1$, generated by the 0-dimensional cochain, sending $p \mapsto 1$. As

$$\alpha(p, q) = p + q$$

α is surjective. Thus, we have

$$0 \longrightarrow \mathrm{H}^1(S^n) \longrightarrow 0$$

and so $\mathrm{H}^1(S^n) = 0$. □

Corollary 2.26. $\mathbb{R}^n$ is homeomorphic to $\mathbb{R}^m$ if and only if $m = n$.

Proof. Suppose $f : \mathbb{R}^n \to \mathbb{R}^m$ is a homeomorphism. This gives us a homeomorphism

$$f : \mathbb{R}^n \setminus 0 \to \mathbb{R}^m \setminus \{f(0)\}$$

But $\mathbb{R}^n \setminus \{*\}$ is homotopy equivalent to $\S^{n-1}$. Thus f induces an isomorphism in homology

$$\mathrm{H}_i(S^{n-1}) = \mathrm{H}_i(S^{m-1})$$

But by our homology computations, this implies that $n = m$. □

Note in the below we assume $n > 0$.

Definition 2.27 (degree)
Suppose $f : S^n \to S^n$ is a continuous map, then f induces a map

$$f_* : \mathrm{H}_n(S^n) \to \mathrm{H}_n(S^n)$$

which is multiplication by some $d \in \mathbb{Z}$. We call $\deg(f) = d$ the *degree* of f.

Remark 2.28. Since $\mathrm{H}_n(S^n) = \mathbb{Z}$ which is free, therefore f_* is determined by $f_*(1)$. Moreover, we will need to use the *same* identification of $\mathrm{H}_n(S^n) = \mathbb{Z}$, and so $\deg(f)$ is well defined. Note that if we use different isomorphisms, then $\deg(f)$ is only defined up to a sign.

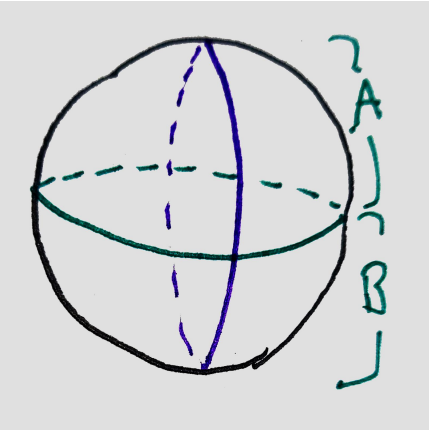
Note that

- if $f \simeq g$, then $\deg(f) = \deg(g)$,
- $\deg(\mathrm{id}) = 1$.
- $\deg(\mathrm{const}) = 0$, since the constant map factors through $\{*\}$, and so the induced map on homology factors through $\mathrm{H}^n(*) = 0$.

Lemma 2.29. If $A \in O(n+1)$, then A acts on S^n, and we have

$$\deg(A) = \det(A) \in \{\pm 1\}$$

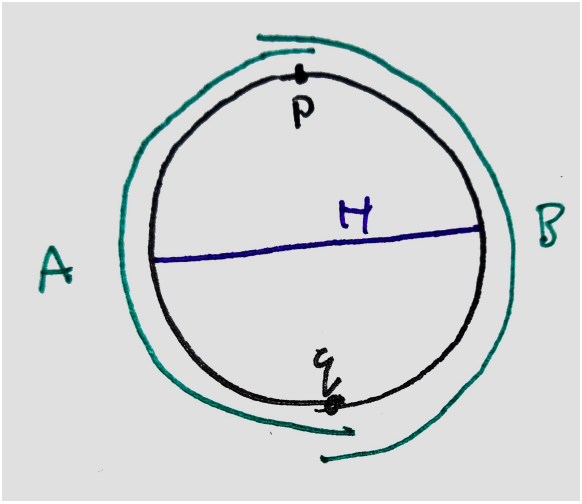
Proof. The group $O(n+1)$ has two path components. By homotopy invariance of degree, any A is homotopic to id_{S^n}, or a reflection in an equatorial hyperplane H.



The reflection preserves $A, B, A \cap B$, and H intersects $A \cap B = S^{n-1}$ at two points. The Mayer-Vietoris sequence gives

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{H}_n(S^n) & \xrightarrow{\cong} & \mathrm{H}_{n-1}(S^{n-1}) & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle A_*} & & \downarrow{\scriptstyle A_*} & & \\ 0 & \longrightarrow & \mathrm{H}_n(S^n) & \xrightarrow[\cong]{} & \mathrm{H}_{n-1}(S^{n-1}) & \longrightarrow & 0 \end{array}$$

The two vertical maps are the same. Therefore it suffices to consider the result on S^1.



Recall our computation of $H_1(S^1)$ has

$$0 \longrightarrow H_1(S^1) \longrightarrow H_0(p \sqcup q) \xrightarrow{\alpha} H_0(A) \oplus H_0(B) \longrightarrow \cdots$$

where $\alpha(u, v) = (u + v, u + v)$. Hence $H_1(S^1)$ is generated by $(1, -1)$, which we can think of as $p - q$. But in this case, reflection in H swaps p and q, and so it acts on $H^1(S^1)$ by multiplication by -1. □

Corollary 2.30. The antipodal map $a_n : S^n \to S^n$ has degree $(-1)^{n+1}$.

Proof. a_n is orthogonal. Or equivalently, it is a composition of $n + 1$ reflections, and we have that

$$\deg(fg) = \deg(f)\deg(g)$$

□

Corollary 2.31. If $f : S^n \to S^n$ has no fixed point, then f is homotopic to the antipodal map.

Proof. In fact, we will see that $f(x) \neq g(x)$ for all $x \in S^n$, then $f \simeq a_n \circ g$. Taking $g = \mathrm{id}$ gives the result. Consider the map

$$x \mapsto \frac{tf(x) - (1-t)g(x)}{\|tf(x) - (1-t)g(x)\|}$$

for $0 \leq t \leq 1$. Note that the denominator never vanishes. This then defines a homotopy from f to $-g$. □

Using this, the degree of any map without fixed points is $(-1)^{n+1}$. Another corollary is that if $f(x) \neq -x$ for all $x \in S^n$, then $f \simeq a \circ a = \mathrm{id}$.

Definition 2.32 (vector field on S^n)
A *vector field on* S^n is a map $v : S^n \to \mathbb{R}^{n+1}$, such that for all $x \in S^n$,

$$\langle x, v(x) \rangle = 0$$

Proposition 2.33 (Hairy ball theorem)**.** S^n has a nowhere vanishing vector field if and only if n is odd.

Proof. If $n = 2k - 1$,

$$v(x_1, y_1, \ldots, x_k, y_k) = (-y_1, x_2, \ldots, -y_k, x_k)$$

works. □

Index

boundary, 3
boundary homomorphism, 2

chain complex, 2
chain map, 3
coboundary, 3
cochain complex, 2
cochain maps, 3
cocycle, 3
cohomology class, 3
cohomology group, 3
k-connected, 2
contractible, 1
cycle, 3

degree
    of a map $S^n \to S^n$, 12
differential, 3

exact at, 8
exact sequence, 8

homology class, 3
homology group, 2
homotopic, 1
homotopy equivalence, 1
homotopy invariance
    of (co)homology, 7

Mayer-Vietoris, 8
    homomorphism, 8

path connected, 2

short exact sequence, 8
simply connected, 2
singular chain complex, 4
singular cochain complex, 5
singular cohomology, 5
singular homology, 5
singular simplex, 4
standard simplex, 4

vector field on S^n, 13